

Question	Marking Guidance	Mark	Comments
09.1	Directional;	1 (1 x AO1)	

Question	Marking Guidance	Mark	Comments
09.2	1. (Use of) penicillin/antibiotic creates selection pressure OR Penicillin/antibiotic kills susceptible/non-resistant <i>S. aureus</i>/bacteria; 2. (Resistant <i>S. aureus</i>/bacteria) are more likely to survive and reproduce to pass on <i>blaZ</i>/penicillin-resistance/beta lactamase allele; 3. Increased frequency of (<i>blaZ</i>) allele (in s offspring/next generation);	3 (1 x AO1, 2 x AO2)	2 and 3. Reject 'gene' for 'allele' once

Question	Marking Guidance	Mark	Comments
09.3	1. (PRSA) can be resistant to many antibiotics OR Many antibiotics destroyed (by PRSA) OR Many antibiotics ineffective (against PRSA); 2. Bacteria would be able to synthesise cell walls and continue reproducing; 3. New/different antibiotics needed OR No new antibiotics in recent years;	2 max (2 x AO2)	1. Accept 'most' for many

Question	Marking Guidance	Mark	Comments
09.4	Correct answer, of $0.19 / 0.2 = 2$ marks;; Incorrect but shows 1.19596 and 1.3875 (accept any correct rounding) = 1 mark OR Incorrect but shows $675 \div (5.644 \times 10^7 \div 100\ 000)$ and $787 \div (5.672 \times 10^7 \div 100\ 000) = 1$ mark;	2 (2 x AO2)	

Question	Marking Guidance	Mark	Comments
09.5	1. Weaker immune systems so more susceptible OR More likely to be infected as close contact with large numbers of people; 2. More likely to be infected as close contact with large numbers of people; 3. More MRSA sufferers so more likely to come into contact with it;	2 max (2 x AO2)	